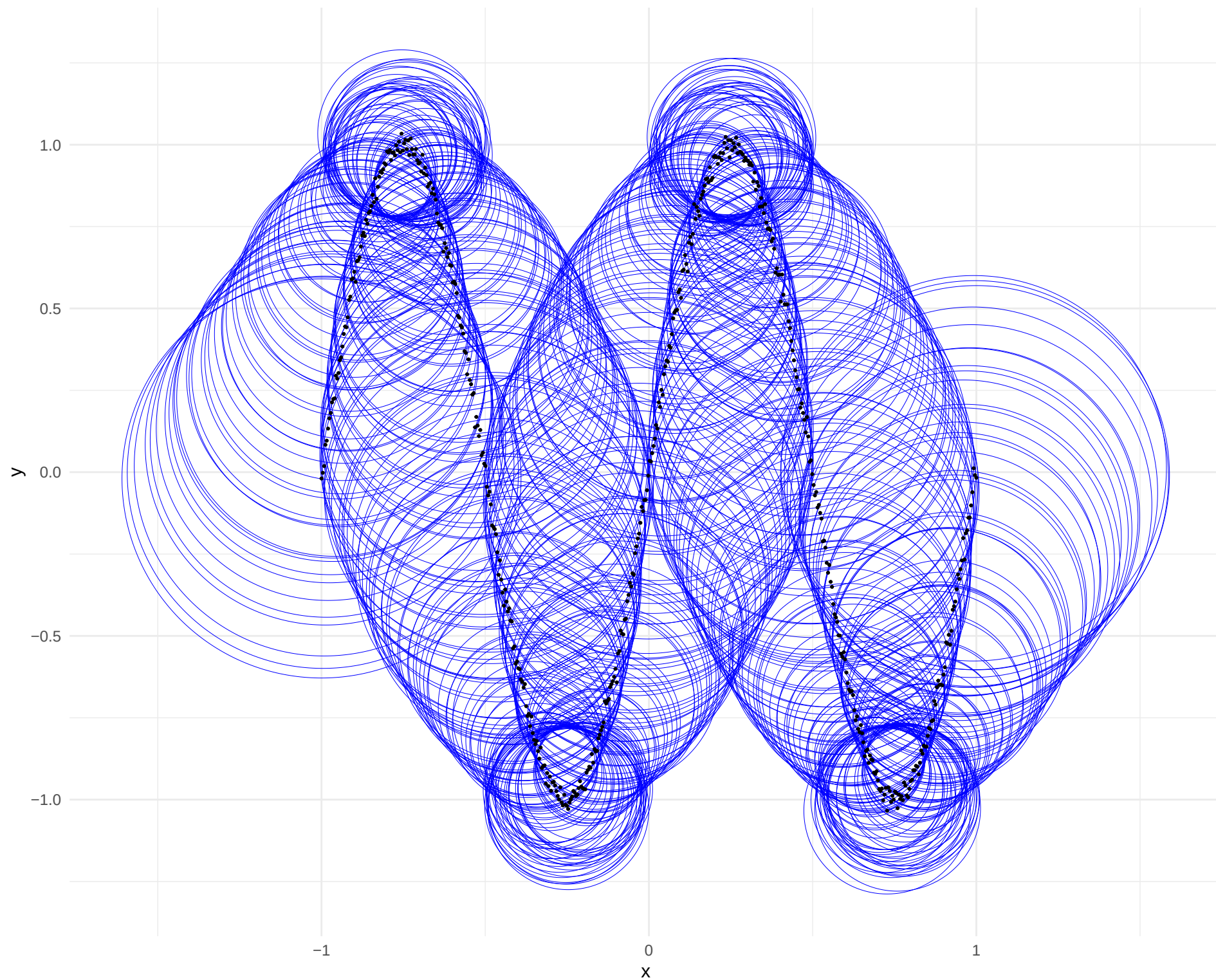
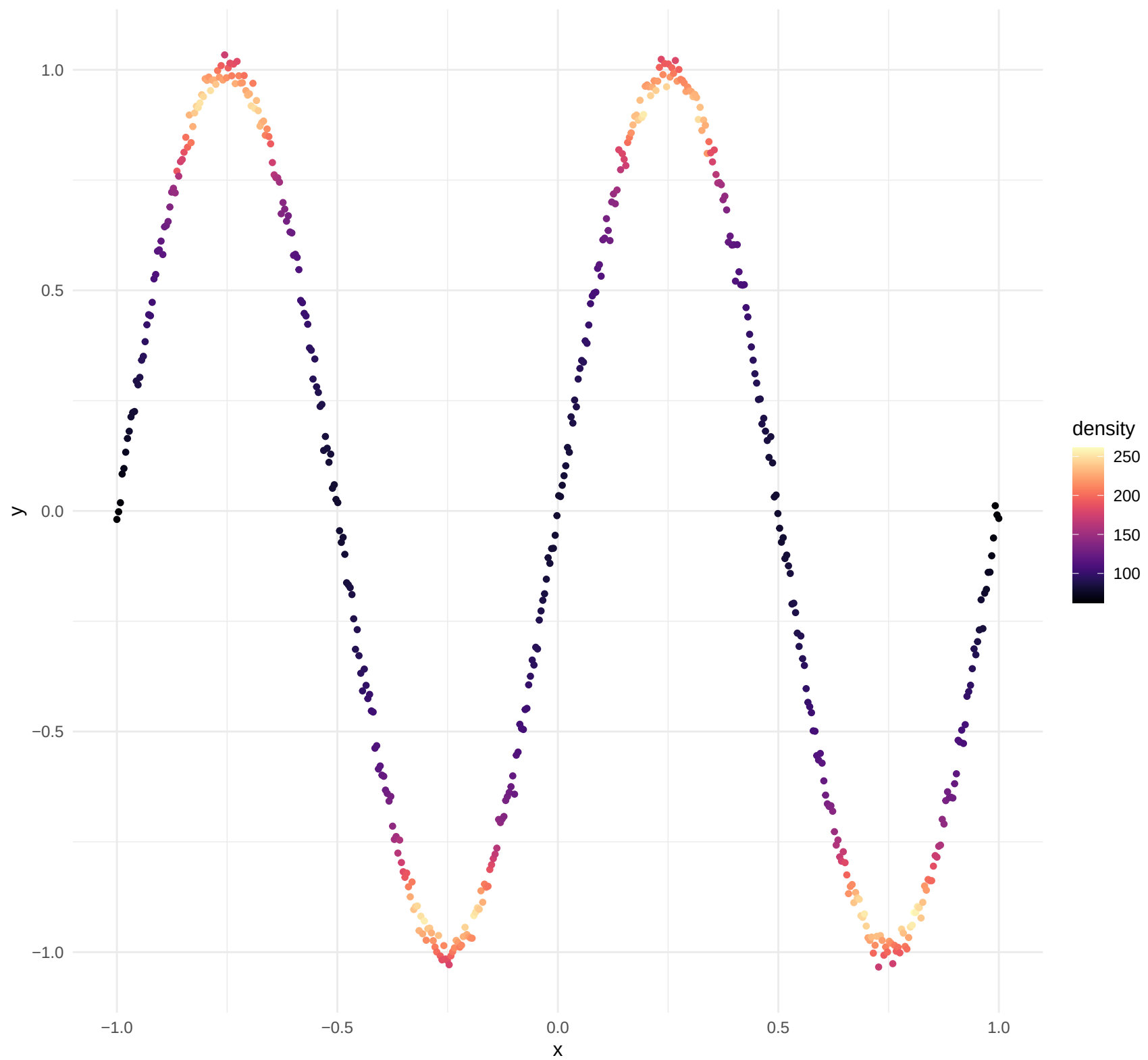


All Adaptive Spheres

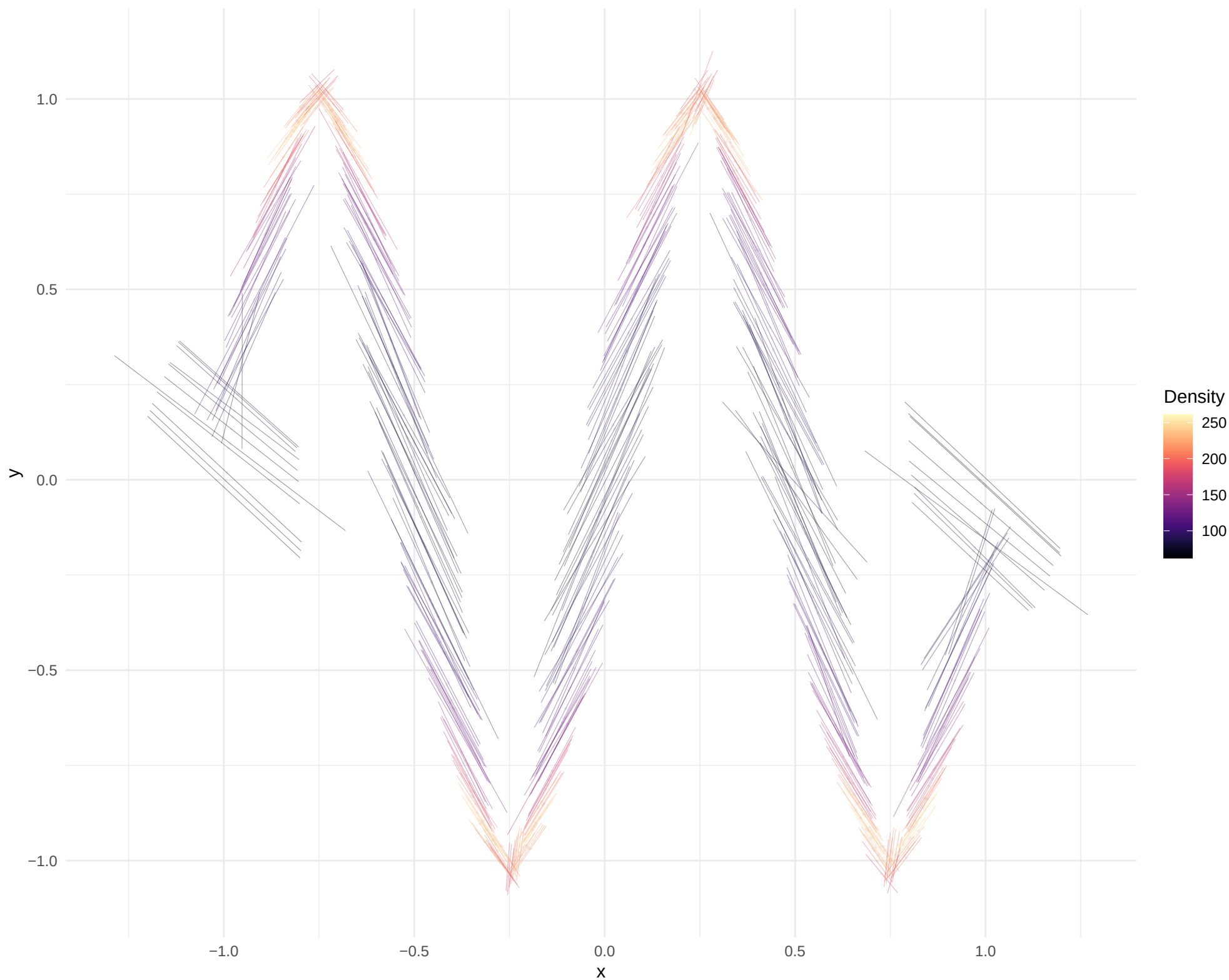
Visualizing local neighborhood size (radius) for every point.



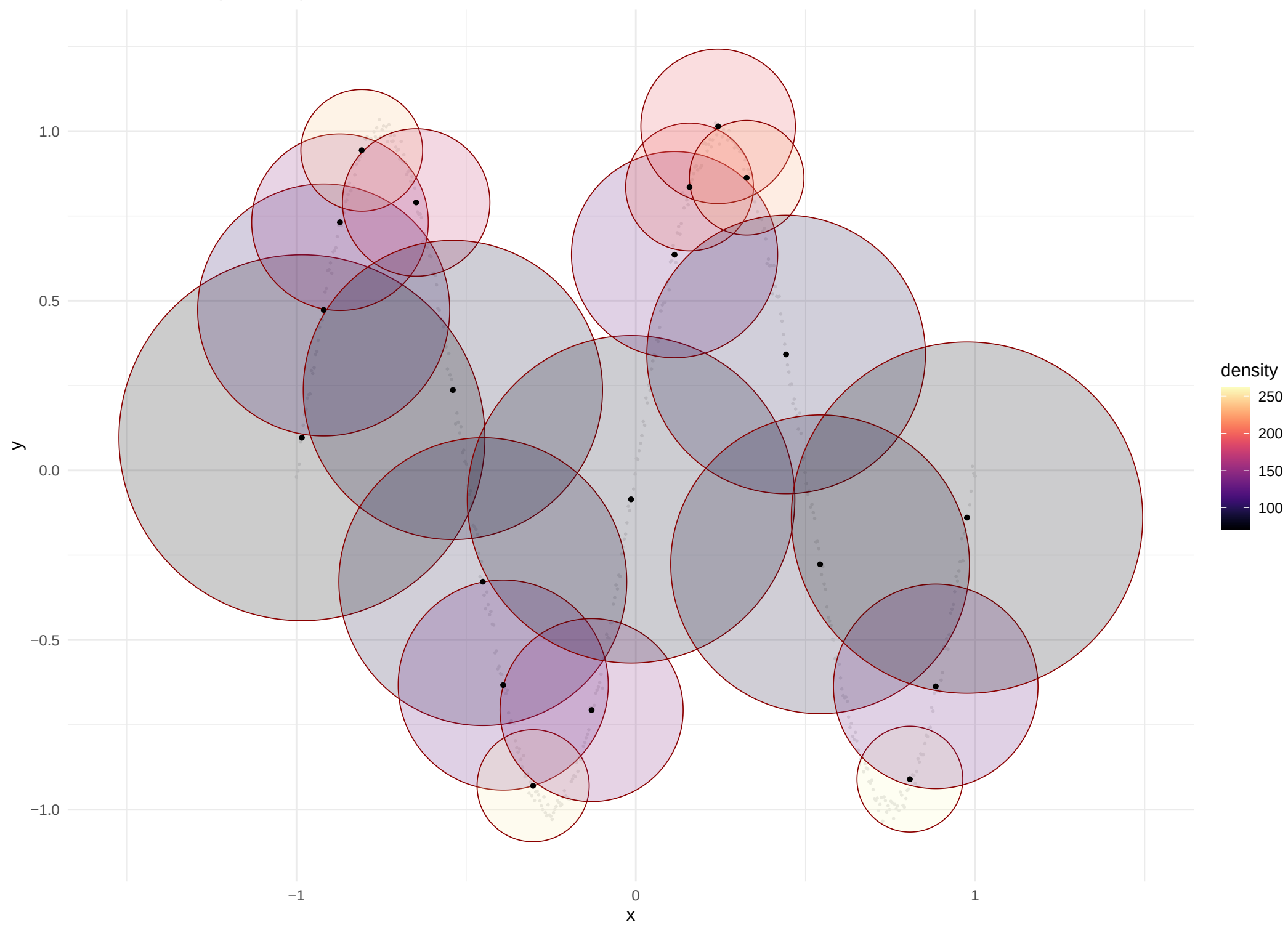
Point Density Distribution



Global Tangent Field (PC1)

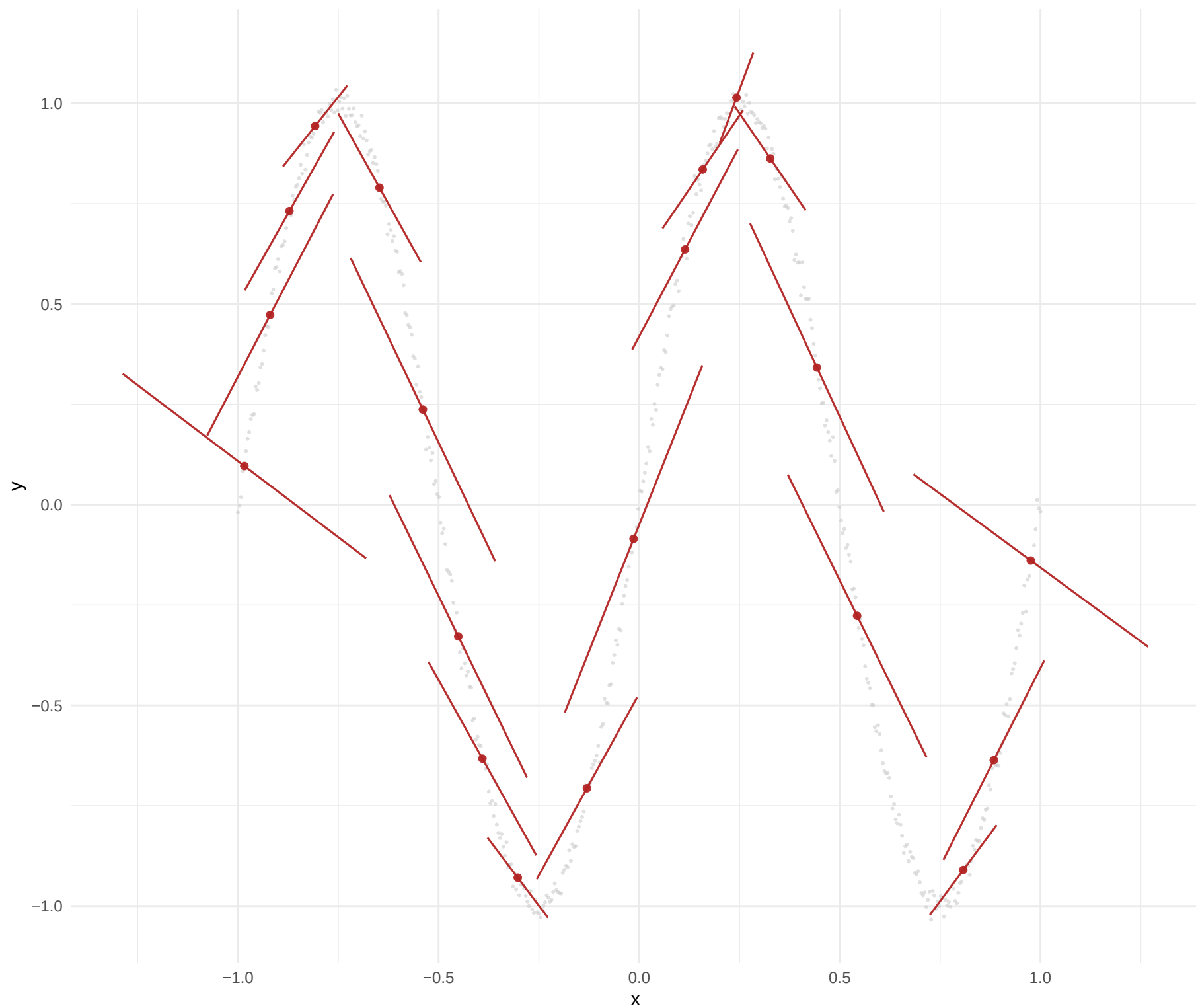


Atlas Selection (Anchors)



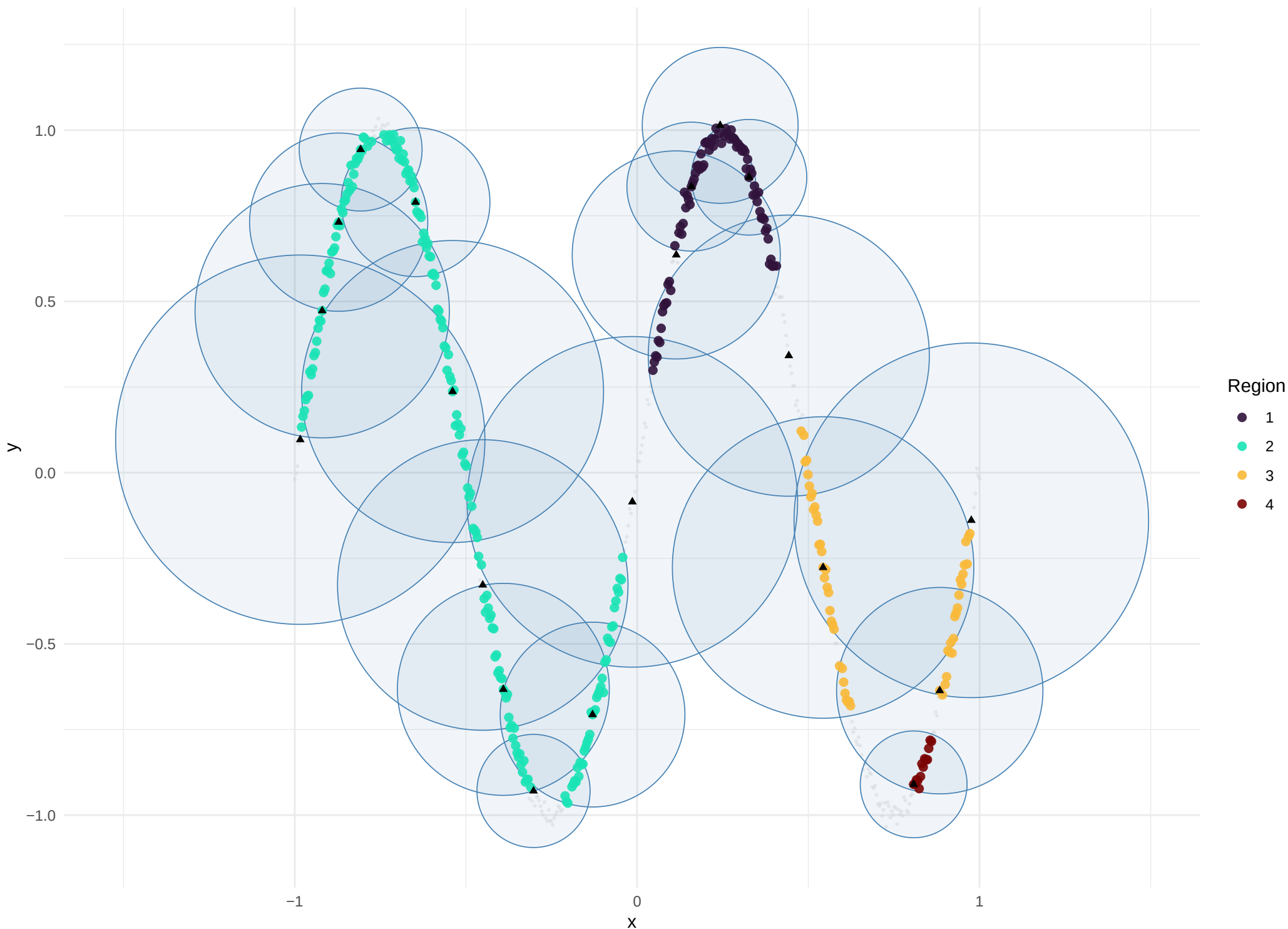
Anchor SVD Segments

Same anchors as plot 4: spheres first, then their SVD (PC1) in red



Intersection Regions (2+ anchors)

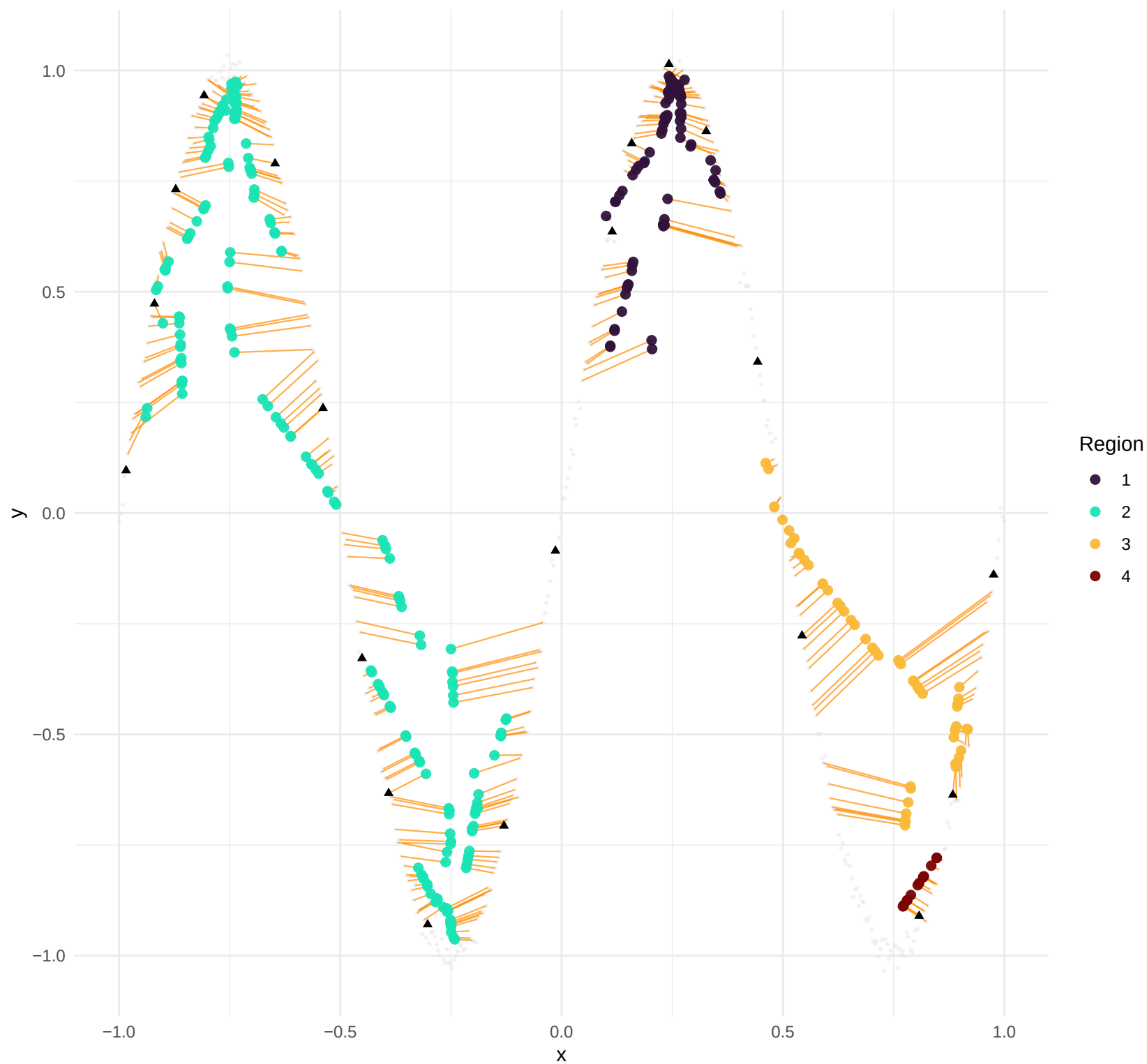
Anchor spheres in blue. Colored points = overlap zones. Triangles = anchor centers.



Parameters: $k_{\min} = 50$, $g_{\text{threshold}} = 3.0$, $o_{\max} = 0.3$, $o_{\min} = 0.10$, $\text{stability} = 0.90$, $\text{scale_factor} = 2.5$, $\text{max_iter} = 50$, $\text{conv_tol} = 0.01$

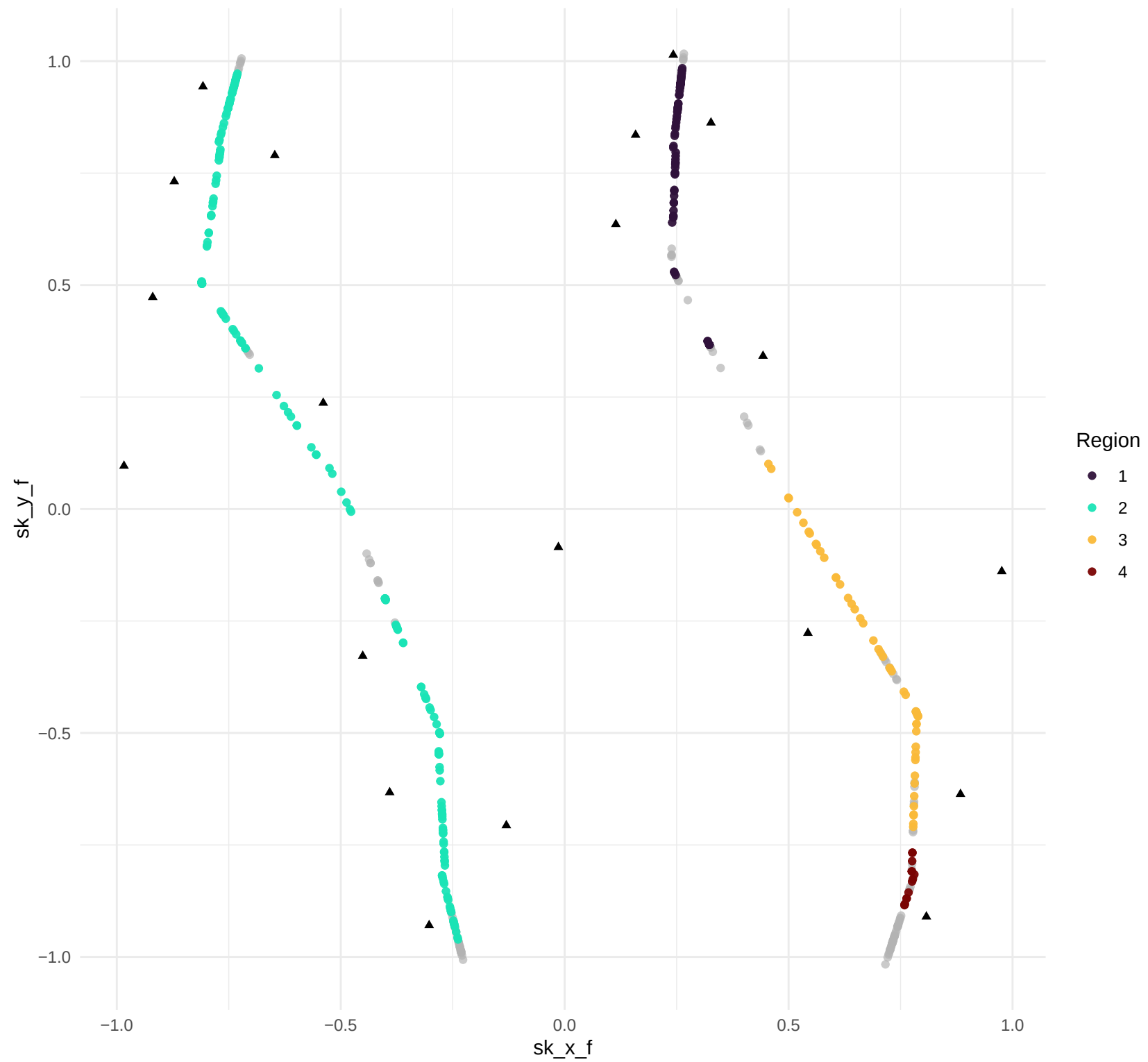
Stitched Backbone (Iteration 1)

Orange path: stitching; Triangles: original anchors



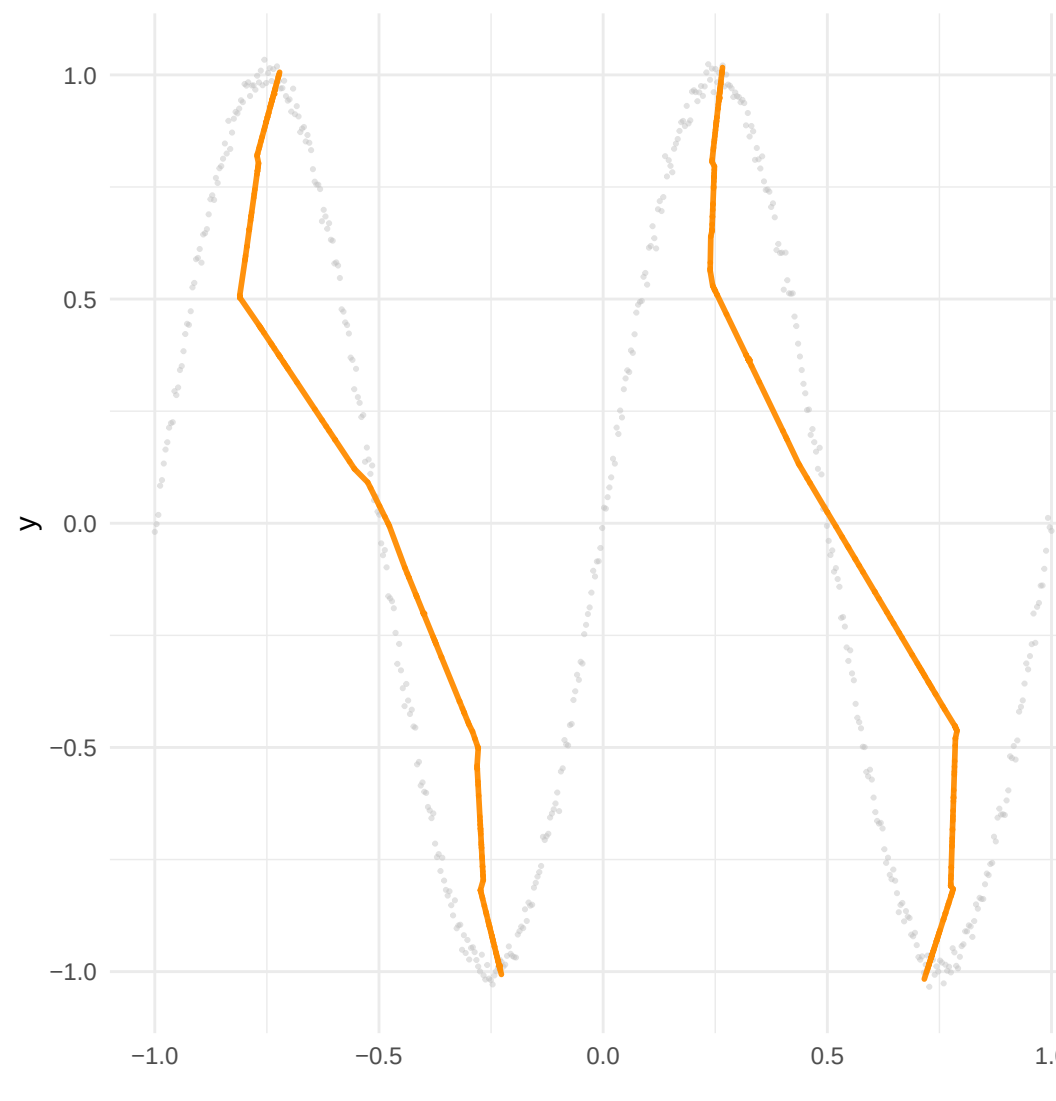
Final Stitched Backbone (Converged)

Final stitched points



Backbone Edges

Final

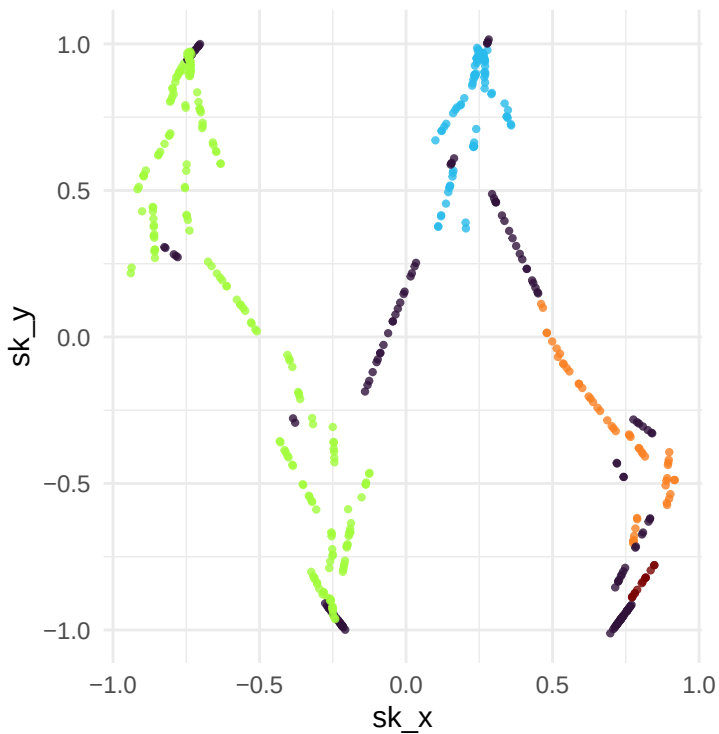


Iteration 1

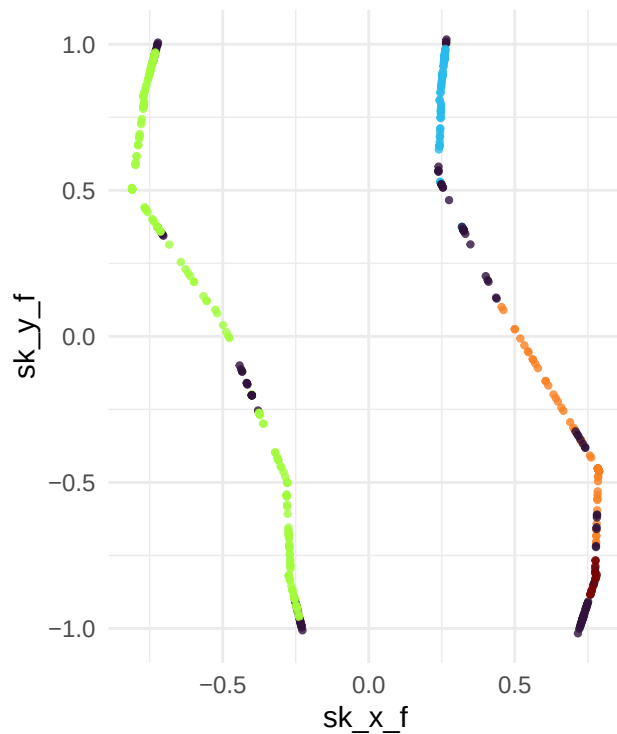


Convergence: iteration 1 vs final

Iteration 1

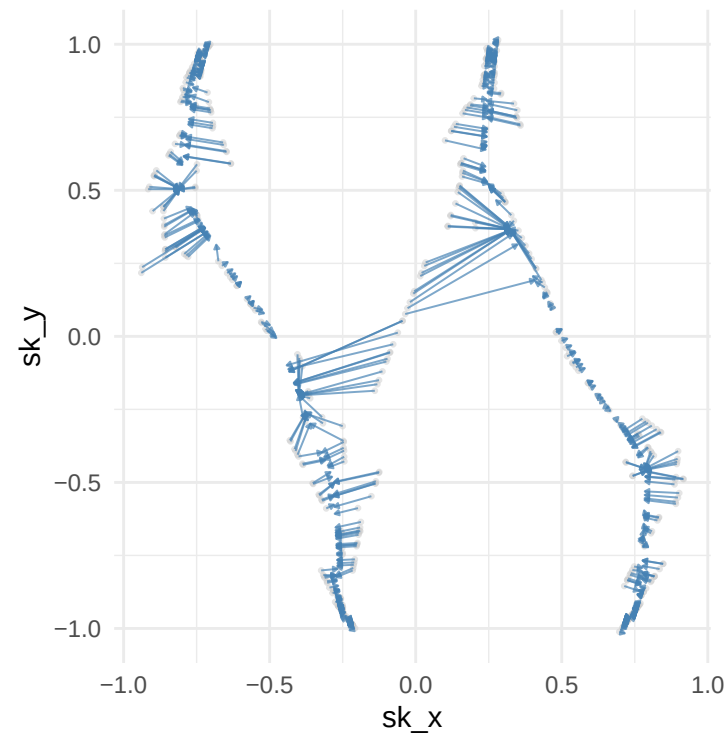


Converged



Displacement iter1 -> final

Points moved > 1e-6: 500 / 500



Parameters: k_min = 50, g_threshold = 3.0, o_max = 0.3, o_min = 0.10, stability = 0.90, scale_factor = 2.5, max_iter = 50, conv_tol = 0.01